

1. Administrative & Study Identification

Full Study Title: KEYMAKER-U01 Substudy 01A: Efficacy and Safety Study of Pembrolizumab (MK-3475) With or Without Chemotherapy When Used With Investigational Agents in Treatment-naïve Participants With Stage IV Non-small Cell Lung Cancer (NSCLC) (MK-3475-01A/KEYMAKER-U01A) **Short Title/Acronym:** KEYMAKER-U01A / MK-3475-01A **Protocol Number:** MK-3475-01A **NCT Number:** NCT04165070 **Posting ID:** 440898076025262280 **Sponsor Name:** Merck **Trial Status:** RECRUITING **Investigational Product:** Pembrolizumab (MK-3475) with or without chemotherapy in combination with investigational agents (e.g., vibostolimab, boserolimab, MK-4830, MK-0482, I-DXd, or HER3-DXd) **Phase of Development:** PHASE1, PHASE2 **Medical Monitor:** Medical Monitor, Merck

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the efficacy and safety of pembrolizumab (MK-3475) with or without chemotherapy in combination with various investigational agents in treatment-naïve participants with advanced squamous or non-squamous NSCLC. **Secondary Objectives:** To evaluate Progression-Free Survival (PFS) and safety measures including AEs and discontinuations due to AEs in specific study parts.

2.2 Background & Rationale Pembrolizumab plus chemotherapy is a standard-of-care first-line treatment for metastatic NSCLC with no targetable genetic alterations. However, there remains an unmet need for patients who do not respond to this standard treatment. Preclinical and early clinical data suggest that combining an immune checkpoint inhibitor with novel investigational agents may provide robust antitumor activity and improve clinical outcomes. This study is one of the pembrolizumab substudies being conducted under one pembrolizumab umbrella master protocol (MK-3475-U01/KEYMAKER-U01 - NCT04165798).

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional (Umbrella Master Study Substudy) **Control Type:** Single-arm (Umbrella platform design evaluating rolling arms) **Blinding:** Open-label **Randomization:** Rolling arms allocation for the different investigational agent combinations

3.2 Planned Enrollment & Duration Target Enrollment: 450 participants **Number of Sites:** Global, multi-center study

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age $\geq$ 18 years with previously untreated histologically or cytologically confirmed Stage IV squamous or nonsquamous NSCLC. 2. Participants with nonsquamous NSCLC who are not eligible for an approved targeted therapy. 3. Is able to provide an archival tumor tissue sample (collected within 5 years or before screening) or newly obtained core or excisional biopsy (within 90 days). 4. Has not received prior systemic treatment for their metastatic NSCLC. 5. Is able to complete all screening procedures within the 35-day window (Part A) or 28-day window (Part B).

4.2 Exclusion Criteria 1. Tumors harboring EGFR, ALK, or ROS1 mutations for which first-line targeted therapy is indicated; or diagnosis of small cell lung cancer. 2. Known history of active Hepatitis B or C infection (HBsAg positive patients are eligible if on antiviral therapy for at least 4 weeks and viral load is undetectable), or HIV (well-controlled HIV with ART is permitted). 3. Prior radiation therapy to the lung >30 Gray (Gy) within 6 months of the first dose, or receipt of a live vaccine within 30 days before the first dose. Any licensed COVID-19 mRNA, adenoviral, or inactivated vaccine is allowed. 4. Active autoimmune disease requiring systemic treatment in the past 2 years, or history of noninfectious pneumonitis requiring steroids. 5. Clinically significant cardiac disease, active CNS metastases, or carcinomatous meningitis.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: On Day 1 of each 3-week cycle (Q3W), participants receive Pembrolizumab 200 mg IV, plus an assigned investigational agent (e.g., vibostolimab (MK-7684), boserolimab (MK-5890), MK-4830, MK-0482, I-DXd, or HER3-DXd) in escalating or defined doses, with or without chemotherapy depending on the assigned rolling arm. **Route of Administration:** Intravenous (IV) **Drug Details:** Includes agents like **carboplatin** (Trade Name: Paraplatin, ATC Code: L01XA02), paclitaxel, and pemetrexed. **Duration of Treatment:** Up to 35 cycles (approximately 2 years) for pembrolizumab; investigational agents can be received until progressive disease or unacceptable toxicity.

5.2 Concomitant & Prohibited Therapy Permitted: Licensed COVID-19 vaccines (mRNA, adenoviral, or inactivated). **Prohibited:** Investigational vaccines, live vaccines within 30 days of the first dose, and concurrent antineoplastic therapy.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Up to 35 Days (Part A) / 28 Days (Part B)	Informed Consent, Tumor Biopsy (archival/new), Baseline Imaging, Labs
Baseline	Day 1	First Dose of Treatment, Pharmacokinetics (PK), Vital Signs
Interim	Q3W (Every 3 Weeks)	Safety Labs, Efficacy Assessment (Tumor Imaging every 6-9 weeks)
End of Study	Up to 35 Cycles	Off-treatment safety follow-up up to 90 days, Survival Follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes

- Part A:** Objective Response Rate (ORR) According to Response Evaluation Criteria in Solid Tumors Version 1.1 (RECIST 1.1) (defined as % of participants with Complete Response or Partial Response).
- Part B:** Number of Participants Who Experience One or More Adverse Events (AEs).
- Part B:** Number of Participants Who Discontinue Study Intervention Due to an Adverse Event (AE).

7.2 Secondary Outcomes

- Part A:** Progression-Free Survival (PFS) According to RECIST 1.1.
- Part A:** Number of Participants Who Experience One or More AEs.
- Part A:** Number of Participants Who Discontinue Study Treatment Due to an AE.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuously collected. An AE is defined as any unfavorable and unintended sign, symptom, disease, or worsening of preexisting condition temporally associated with study treatment and irrespective of causality.

Serious Adverse Events (SAEs): Monitored and documented up to 90 days after the last treatment.

Data Monitoring Committee (DMC): Internal sponsor safety review committees evaluate rolling arm dose-escalation and safety profiles continuously.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set for all AE/SAE evaluations and DLT incidence; Efficacy Analysis Set (Intent-to-Treat) for ORR and PFS endpoints.

Sample Size Justification: Total targeted enrollment of 450 participants powered adequately for a Phase 1/2 rolling-arm umbrella design to identify safety signals and preliminary efficacy within specific NSCLC cohorts.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Standard clinical onsite and remote monitoring to ensure protocol adherence and safety tracking. **Audit Trail:** Independent verification of tumor response via RECIST 1.1 and comprehensive eCRF locking mechanisms.

11. Study Identifiers & Governance

Primary ID: MK-3475-01A **Registry Number:** NCT04165070 **Posting ID:** 440898076025262280 **Master Protocol:** MK-3475-U01 (KEYMAKER-U01) - NCT04165798 **Sponsor/Lead Organization:** Merck **Study Title:** KEYMAKER-U01 Substudy 01A **Preferred UMLS Name:** Immunotherapy for Non-Small Cell Lung Carcinoma **Official Regulatory Title:** KEYMAKER-U01 Substudy 01A: A Phase 1/2, Umbrella Study With Rolling Arms of Investigational Agents With Pembrolizumab With or Without Chemotherapy in Treatment-Naive Participants With Stage IV Non-small Cell Lung Cancer (NSCLC)

12. Clinical Framework (Disease Characteristics)

Disease Areas: Carcinoma, Non-Small-Cell Lung **Preferred Name:** Carcinoma **Definition:** A malignant tumor arising from epithelial cells. Carcinomas that arise from glandular epithelium are called adenocarcinomas, those that arise from squamous epithelium are called squamous cell carcinomas, and those that arise from transitional epithelium are called transitional cell carcinomas. **Semantic Type:** Neoplastic Process **Ontology Codes:** * **MeSH Code:** D002277 * **HPO Code:** HP:0030731 * **SNOMED ID:** 1187425009 **Medical Methodology:** Anti-PD-1 immunotherapy combined with novel Investigational Agents and Chemotherapy. **Optimization Target:** Identifying tolerable, robust first-line antitumor combination regimens for treatment-naïve patients.

13. Study Procedures & Methodology

Management Pathway: Master screening protocol (KEYMAKER-U01 - NCT04165798) allocating patients to specific rolling substudies.

Education Protocol (HCP): Protocol training on identifying and managing immune-related AEs (irAEs) and specific related toxicities. **Education Protocol (Patient):** Thorough informed consent regarding the umbrella design, combination treatment schedule, and symptom self-reporting. **Quality Audit Mechanism:** Independent imaging and tumor response verification.

14. Observation & Follow-Up Schedule

Total Duration: Follow-up tracking extending until estimated completion. **Onsite Visit Cadence:** Every 3 weeks (Day 1 of each 3-week cycle) for dosing and evaluation. **Remote Monitoring Frequency:** Continuous safety and AE follow-up up to 90 days post-treatment. **Checklist Review Interval:** Radiological tumor assessments via CT/MRI every 6 to 9 weeks per RECIST v1.1.

15. Sign-off & Approval

Role	Name	Signature	Date		----	----	----	----
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					